

Fluid Mechanics

BRANCH: MECHANICAL ENGG. (ME)

COURSE CODE: ME203

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Fluid Mechanics (ME203) under the Mechanical Engg. (ME) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Fluid Properties & Statics

Viscosity, surface tension, capillarity. Newton's Law of Viscosity. Fluid statics: pressure measurement, manometers, hydrostatic forces on submerged surfaces, buoyancy, metacentric height.

Unit 2: Fluid Kinematics & Dynamics

Lagrangian vs Eulerian descriptions, velocity, acceleration, stream line, path line, streak line, continuity equation. Euler's and Bernoulli's equation, applications: Venturimeter, Orificemeter.

Unit 3: Boundary Layer & Pipe Flow

Laminar vs Turbulent flow, Reynolds experiment. Major and minor losses in pipes. Boundary layer thickness, displacement, momentum thickness, drag and lift.

Unit 4: Fluid Machines

Impact of jets on stationary and moving plates. Turbines: Pelton, Francis, Kaplan (working principles). Centrifugal and Reciprocating pumps.

Bernoulli's Equation: $\frac{P}{\rho} + g + \frac{v^2}{2g} + z = \text{constant}$. Reynolds Number: $Re = \frac{\rho \cdot v \cdot D}{\mu}$.

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



